IGCSE 0607 Practice Questions — E5.2

Angle measurement in degrees

Total Questions: 12 — **Total Marks:** 69 Marks

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- **E5.2** Angle measurement in degrees
  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 The positions of three towns, A, B and C, are marked on a map.

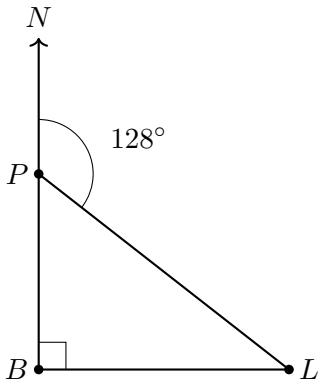
(a) The bearing of town B from town A is 068°.
Work out the bearing of town A from town B.

..... ° **[2]**

(b) Town C is due west of town B.
Write down the bearing of town C from town B.

..... ° **[1]**

2 The bearing of a lighthouse L from a port P is 128°.
A boat B is positioned such that it is due South of P and due West of L.



(a) Find the bearing of P from L.

..... ° **[2]**

(b) Work out the size of angle BPL.

..... ° **[1]**

(c) Work out the size of angle PLB.

..... ° **[2]**

3 The bearing of a lighthouse, L, from a boat, B, is 148°.

(a) Work out the bearing of B from L.

..... ° **[2]**

(b) A port, P, is due north of L and due east of B.

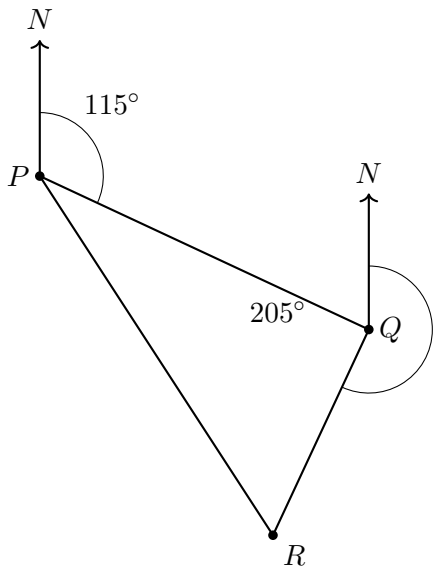
(i) Write down the bearing of B from P.

..... ° **[1]**

(ii) Work out the size of angle LBP.

..... ° **[2]**

4 The diagram shows the positions of three points P, Q and R.
The bearing of Q from P is 115°.
The bearing of R from Q is 205°.



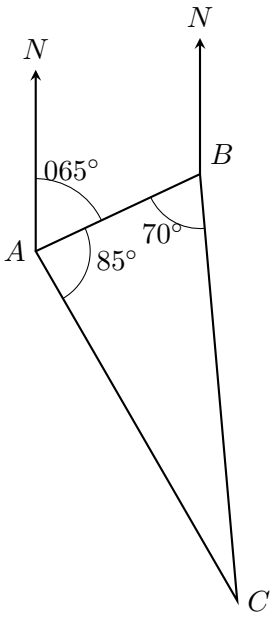
(a) Show that angle $PQR = 90°$.

[2]

(b) The distance PQ is 15 km and the distance QR is 8 km.
Calculate the distance PR.

..... km **[3]**

5 The diagram shows the positions of three towns, A, B and C.
The bearing of B from A is 065°.
Angle $CAB = 85°$ and angle $ABC = 70°$.



(a) Calculate the bearing of C from A.

..... ° **[1]**

(b) Calculate the bearing of A from B.

..... ° **[2]**

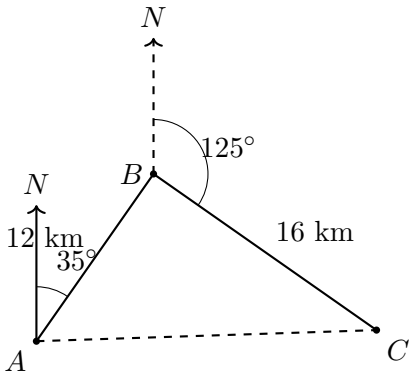
(c) Calculate the bearing of C from B.

..... ° **[2]**

(d) Calculate angle ACB.

..... ° **[1]**

6 A ship leaves a port A and travels 12 km on a bearing of 035° to point B.
From B, the ship travels 16 km on a bearing of 125° to point C.



(a) Show that angle $ABC = 90°$.

[2]

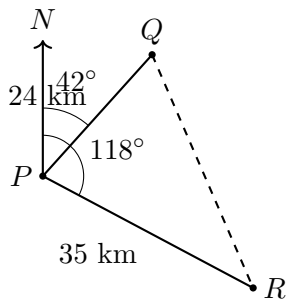
(b) Calculate the direct distance AC.

$AC =$ ..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[3]**

7 Three towns are located at points P, Q, and R.
Town Q is 24 km from P on a bearing of 042°.
Town R is 35 km from P on a bearing of 118°.



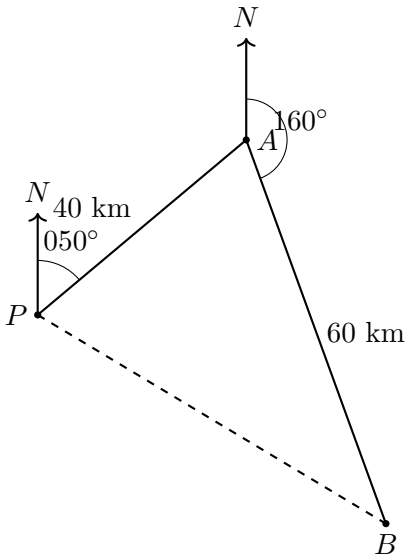
(a) Calculate the distance QR.

$QR =$ ..... km **[3]**

(b) Calculate the bearing of R from Q.

..... ° **[4]**

8 A ship leaves port P and sails 40 km on a bearing of 050° to reach point A.
It then sails 60 km on a bearing of 160° to reach point B.



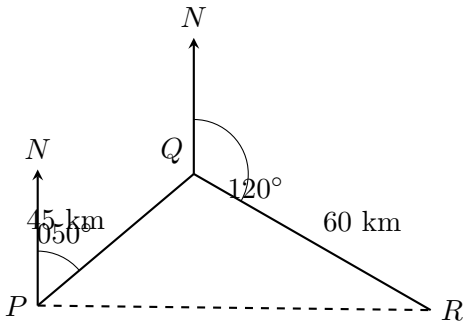
(a) Calculate the distance PB.

..... km **[4]**

(b) Calculate the bearing of B from P.

..... ° **[4]**

9 A ship sails 45 km from Port P on a bearing of 050° to Point Q.
It then sails 60 km on a bearing of 120° to Point R.



(a) Calculate the distance PR.

..... km **[4]**

(b) Calculate the bearing of R from P.

..... ° **[3]**

(c) The ship sails directly back from R to P.
Calculate the bearing of P from R.

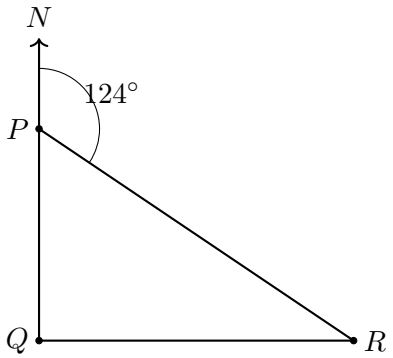
..... ° **[2]**

10 The bearing of point Y from point X is 048°.

Work out the bearing of point X from point Y.

..... ° **[2]**

11 The diagram shows the positions of three points, P, Q and R.
Point Q is due south of point P.
Point R is due east of point Q.
The bearing of point R from point P is $124°$.



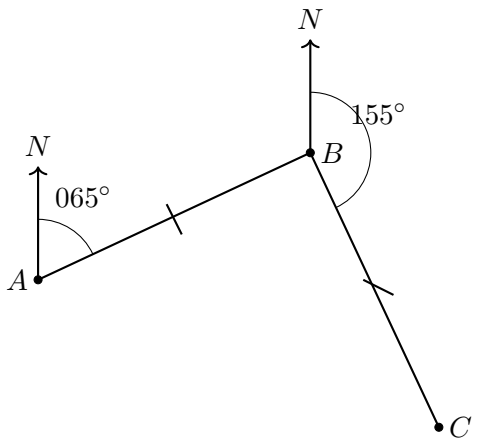
(a) Find the bearing of point P from point R.

..... ° **[2]**

(b) Calculate the angle PRQ.

Angle PRQ = ... ° **[2]**

12 Point B is on a bearing of 065° from point A.
Point C is on a bearing of 155° from point B.
The distance $AB = BC$.



(a) Show that angle $ABC = 90°$.

[2]

(b) Work out the bearing of A from C.

..... ° **[3]**

(c) Point D is such that the quadrilateral $ABCD$ is a square.
Find the bearing of D from A.

..... ° **[2]**